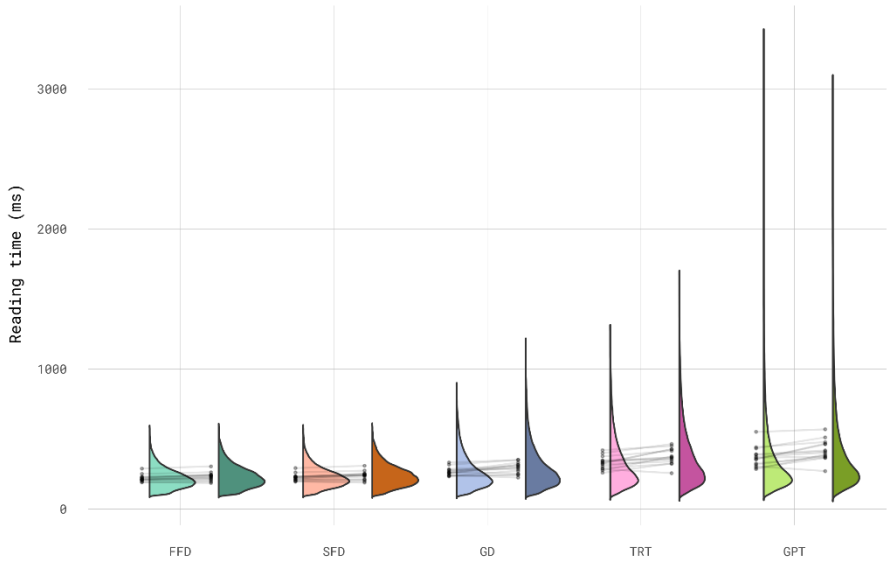
Supplementary Materials

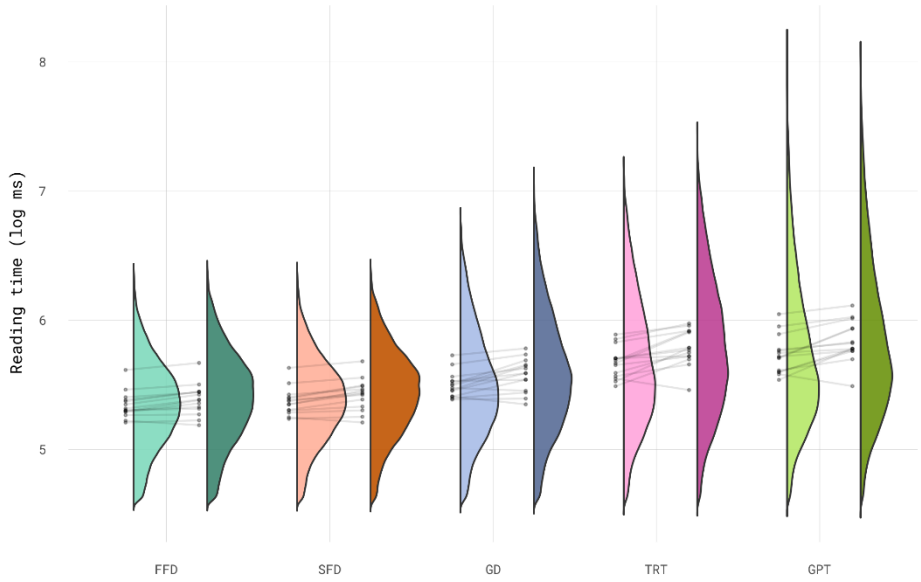
Analyses on the first part of the sample (15 participants)

Figure S1 Distribution of the reading times for L1 and L2

a.



b.



Note: Distribution of the (a) raw and (b) log-transformed reading times for participants with dyslexia reading in their L1 (Dutch) and L2 (English). Points represent individual participants' mean reading time. Distributions of L1 reading times are shown in lighter shades, L2 distributions in darker shades of the same colour.

Table S1

Averages, standard deviation (SD), and maximal values for the reading time measures for L1 (Dutch) and L2 (English) reading of the first 15 participants. Note that participant 1 was excluded because of below chance level score on the comprehension questions. Due to trimming the minimum value is 100 ms for each measure. The result of the t-test comparing L1 and L2 reading in DECO is presented in the t-value column with degrees of freedom in brackets. Statistical significance is indicated by asterisks: $p < .05$ (*), $p < .01$ (**), and $p < .001$ (***).

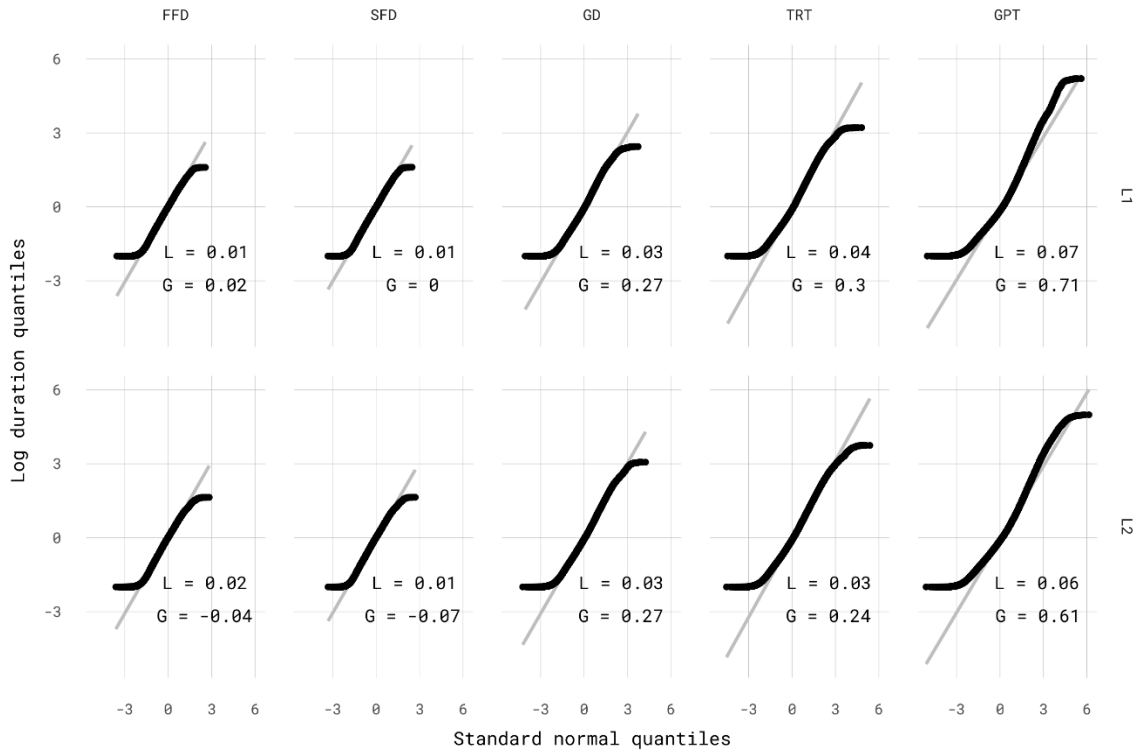
	L1 (Dutch)			L2 (English)			L1 - L2
	Mean	SD	Max	Mean	SD	Max	*t* [df]
First-fixation duration (ms)	220.01	72.89	582	232.24	80.24	593	-4.78 [13]***
Single-fixation duration (ms)	226.64	72.84	584	240.78	79.72	594	-5.05 [13]***
Gaze duration (ms)	266.69	119.39	879	294.90	145.44	1192	-3.51 [13]**
Total reading time (ms)	328.72	177.69	1283	376.69	216.14	1663	-4.59 [13]***
Go past time (ms)	367.34	272.08	3393	417.42	316.56	3057	-4.37 [13]***

Table S2

Comparison of the reading times in the current DECO corpus (first 15 participants) and the previous GECO corpus with 95% confidence intervals (95% CI) and test statistics for both L1 (Dutch) and L2 (English) reading. Positive values mean the reading times were longer in the DECO corpus. Comparisons for which the normality assumption held used Welch's *t*-test (mean difference, *t* [df] reported); comparisons for which normality was violated used the Mann-Whitney *U* test (W* reported). Statistical significance is indicated by asterisks: $p < .05$ (*), $p < .01$ (**), and $p < .001$ (***).

	L1 (Dutch)				L2 (English)			
	Test	Δ	95% CI	Statistic [df]	Test	Δ	95% CI	Statistic [df]
First-fixation duration	Mann-Whitney U	9.09	[-6.43, 26.01]	W = 160	Welch's t	10.65	[-8.63, 29.94]	1.15 [21.2]
Single-fixation duration	Welch's t	16.92	[-0.59, 34.44]	1.99 [24.6]	Welch's t	13.93	[-6.12, 33.98]	1.44 [22.7]
Gaze duration	Welch's t	40.78	[19.91, 61.66]	4.01 [27.2]***	Mann-Whitney U	43.10	[10.97, 73.14]	W = 208**
Total reading time	Welch's t	70.06	[39.88, 100.23]	4.81 [22.5]***	Welch's t	70.37	[33.08, 107.65]	3.9 [23.2]***
Go-past time	Welch's t	79.74	[34.86, 124.62]	3.7 [20.5]**	Welch's t	72.97	[22.5, 123.45]	2.98 [25.4]**

Figure S2 QQ-plots of log-transformed reading times



Note: Quantile-quantile plots of the trimmed and log-transformed reading times. Two statistics are included L (Lilliefors test of normality) and G (Pearson's moment coefficient of skewness). Larger values of L mean stronger deviation of normality and larger values of G mean more skewness in the distribution (positive values mean rightward skewness).

Table S3

Average skipping and refixation probability for L1 (Dutch) and L2 (English) reading for the current DECO corpus (first 15 participants) and the previous GECO corpus. Statistical significance is indicated by asterisks: $p < .05$ (*), $p < .01$ (**), and $p < .001$ (***).

	L1 (Dutch)				L2 (English)				*t* Value (L1–L2)
	Mean	SD	Min	Max	Mean	SD	Min	Max	*t* [df]
DECO									
Skipping probability	0.19	0.06	0.13	0.32	0.16	0.05	0.11	0.29	3.33[13]**
Refixation probability	0.21	0.05	0.14	0.30	0.25	0.05	0.15	0.31	-2.82[13]*
GECO									
Skipping probability	0.39	0.08	0.22	0.52	0.32	0.07	0.18	0.48	8.42[17]***
Refixation probability	0.10	0.04	0.05	0.20	0.15	0.06	0.06	0.29	-8.33[17]***

Table S4

Spearman-Brown split-half reliability coefficients for L1 (Dutch) and L2 (English) reading of the first 15 participants of the DECO corpus.

	L1 (Dutch)	L2 (English)
First-fixation duration	0.48	0.51
Single-fixation duration	0.56	0.61
Gaze duration	0.73	0.75
Total reading time	0.78	0.80
Go past time	0.79	0.80

Split-half reliability

Table S5

Spearman-Brown split-half reliability coefficients for the first and second half of the novel separately as participants read only half of the novel in one language.

	L1 (Dutch)				L2 (English)			
Measure	Reliability first half	Min-Max	Reliability second half	Min-Max	Reliability first half	Min-Max	Reliability second half	Min-Max
First-fixation duration	0.41	0.35-0.44	0.54	0.46-0.58	0.59	0.55-0.61	0.43	0.38-0.46
Single-fixation duration	0.48	0.44-0.51	0.64	0.55-0.69	0.65	0.59-0.68	0.58	0.54-0.6
Gaze duration	0.69	0.67-0.71	0.77	0.75-0.78	0.80	0.78-0.82	0.70	0.66-0.72
Total reading time	0.76	0.75-0.77	0.80	0.78-0.81	0.83	0.82-0.84	0.76	0.75-0.77
Go past time	0.78	0.77-0.79	0.80	0.78-0.81	0.84	0.82-0.85	0.77	0.76-0.78

Variables and Materials

Table S6

Description of the variables included in the datafiles of the DECO corpus.

Variable	Content
PP_NR	The identification number of the participant.
LANGUAGE_RANK	Factor indicating whether the participants read this part in their first language ("L1") or their second ('L2").
LANGUAGE	Factor indicating in which language the current part was read ("Dutch" or "English").
SESSION	Whether the data was collected in the first or the second part of the study.
PART	The number of the part of the novel.
TRIAL	The number of the trial. Starts at 5 as there are 4 practice trials.
TRIAL_FIXATION_COUNT	The total number of fixations in the current trial.
TRIAL_TOTAL_READING_TIME	Summation of all fixation durations in the current trial.
WORD_ID_WITHIN_TRIAL	Chronological identification number of the word within the trial.
WORD_ID	Identification number of the word. The first number refers to the part of the novel (1,2,3 or 4), the second number refers to the trail number, the last number refers to the word number within the trial.
WORD	The word contained in the current interest area.
WORD_AVERAGE_FIX_PUPIL_SIZE	Average pupil size across all fixations in the current word.

WORD_FIXATION_COUNT	Total fixation falling within the current word.
WORD_FIXATION_%	Percentage of all fixations in a trial falling in the current word.
WORD_RUN_COUNT	The number of times the current word was entered and left (runs).
WORD_FIRST_RUN_START_TIME	The start time of the first run of fixations in the current word.
WORD_FIRST_RUN_END_TIME	The end time of the first run of fixations in the current word.
WORD_FIRST_RUN_FIXATION_COUNT	The number of all fixations in a trial falling in the first run of the current word.
WORD_FIRST_RUN_FIXATION_%	Percentage of all fixations in a trial falling in the first run of the current word.
WORD_GAZE_DURATION	Summation of all fixation durations in the first run within the current word.
WORD_SECOND_RUN_START_TIME	The start time of the second run of fixations in the current word.
WORD_SECOND_RUN_END_TIME	The end time of the second run of fixations in the current word.
WORD_SECOND_RUN_FIXATION_COUNT	The number of all fixations in a trial falling in the second run of the current word.
WORD_SECOND_RUN_FIXATION_%	Percentage of all fixations in a trial falling in the second run of the current word.
WORD_SECOND_RUN_DWELL_TIME	Summation of all fixation durations in the second run within the current word.
WORD_THIRD_RUN_START_TIME	The start time of the third run of fixations in the current word.

WORD_THIRD_RUN_END_TIME	The end time of the third run of fixations in the current word.
WORD_THIRD_RUN_FIXATION_COUNT	The number of all fixations in a trial falling in the third run of the current word.
WORD_THIRD_RUN_FIXATION_%	Percentage of all fixations in a trial falling in the third run of the current word.
WORD_THIRD_RUN_DWELL_TIME	Summation of all fixation durations in the third run within the current word.
WORD_FIRST_FIXATION_DURATION	The duration of the first fixation that was within the current word.
WORD_FIRST_FIXATION_INDEX	The ordinal sequence of the first fixation that was within the current word.
WORD_FIRST_FIXATION_RUN_INDEX	The number of runs of fixations have occurred when a first fixation is made to the current word. The current run is included in the tally.
WORD_FIRST_FIXATION_TIME	The start time of the first fixation to enter the current word.
WORD_FIRST_FIXATION_VISITED_WORD_COUNT	The number of different words visited before the first fixation is made into the current word.
WORD_FIRST_FIXATION_X	The horizontal coordinate position of the first fixation that was within the current word.
WORD_FIRST_FIXATION_Y	The vertical coordinate position of the first fixation that was within the current word.
WORD_FIRST_FIX_PROGRESSIVE	Factor indicating whether later words have been visited before the first fixation enters the current word ("0") or not ("1").

WORD_SECOND_FIXATION_DURATION	The duration of the second fixation in the current word, regardless of the run.
WORD_SECOND_FIXATION_RUN	The run index of the second fixation in the current word.
WORD_SECOND_FIXATION_TIME	The time of the second fixation in the current word, regardless of run.
WORD_SECOND_FIXATION_X	The horizontal coordinate position of the second fixation that was within the current word.
WORD_SECOND_FIXATION_Y	The vertical coordinate position of the second fixation that was within the current word.
WORD_THIRD_FIXATION_DURATION	The duration of the third fixation in the current word, regardless of the run.
WORD_THIRD_FIXATION_RUN	The run index of the third fixation in the current word.
WORD_THIRD_FIXATION_TIME	The time of the third fixation in the current word, regardless of run.
WORD_THIRD_FIXATION_X	The horizontal coordinate position of the third fixation that was within the current word.
WORD_THIRD_FIXATION_Y	The vertical coordinate position of the third fixation that was within the current word.
WORD_LAST_FIXATION_DURATION	The duration of the last fixation in the current word, regardless of the run.
WORD_LAST_FIXATION_RUN	The number of all fixations in a trial falling in the last run of the current word.
WORD_LAST_FIXATION_TIME	The time of the last fixation in the current word, regardless of run.
WORD_LAST_FIXATION_X	The horizontal coordinate position of the last fixation that was within the current word.

WORD_LAST_FIXATION_Y	The vertical coordinate position of the last fixation that was within the current word.
WORD_GO_PAST_TIME	Summation of all fixation durations from when the current word is first fixated until the eyes enter a word with a higher word identification number.
WORD_SELECTIVE_GO_PAST_TIME	Summation of all fixation durations in the current word from when the current word is first fixated until the eyes enter a word with a higher word identification number.
WORD_TOTAL_READING_TIME	Summation of all fixation durations in the current word.
WORD_TOTAL_READING_TIME_%	Percentage of trial time spent in the current word.
WORD_SPILLOVER	The duration of the first fixation made in the next word after leaving the current word in the first pass.
WORD_SKIP	A word is considered skipped (i.e., WORD_SKIP = 1) if no fixation occurred in first-pass reading.

Table S7

Description of the language skills measure collected from the participants. Results of these tests can be found in the PP Data folder on OSF.

PP_NR	The identification number of the participant.
AGE	Age of the participant in years.
SEX	Sex of the participant. male/female
AOA_ENG	Age of acquisition of English. The age at which they started learning English.
%EXP_DUTCH	The amount of exposure to Dutch.
%EXP_ENG	The amount of exposure to English.
LEXTALE_DUTCH	Score on the (unspeeded) Dutch lexical decision task: LexTALE; Lemhöfer & Broersma, 2012
LEXTALE_ENG	Score on the (unspeeded) English lexical decision task: LexTALE; Lemhöfer & Broersma, 2012
SPELLING_DUTCH	Score on the Dutch spelling test: GLETSCHR (De Pessemier & Andries, 2009)
SPELLING_ENG	Score on the English spelling subtest of the WRAT 4 (Wilkinson & Robertson, 2006)
COMPR_DUTCH	Percentage of the comprehension questions answered correctly for the parts read in Dutch.
COMPR_ENG	Percentage of the comprehension questions answered correctly for the parts read in English.
LD_Dutch_Acc	Score on a speeded Duch lexical decision task.
LD_English_Acc	Score on a speeded Duch lexical decision task.
ORT	One minute reading test in Dutch. Participants read as many words as possible correctly (Tops et al., 2019).
AGE_DIAGNOSIS	The age at which they got the diagnosis of dyslexia.
SUBJ_SEV	A subjective estimation of the severity of their reading impairment. Light/Mediocre/Severe
DIFF	The area in which they experience the most difficulties. Reading & Spelling/Reading/Spelling

SESSION	Indicates whether the data of the participant was collected in the first or second session.

Comprehension Questions per Chapter

Chapter 1

1. Mr. Hastings takes a sudden dislike to which one of these persons?
   a) John cavendish
   b) Mary Cavendish
   c) Evelyne Howard
   d) Alfred Inglethorpe

2. The book is narrated by Mr. Hastings. Of what secret ambition does he tell Mary Cavendish?
   a) Becoming a biographer.
   b) Becoming a police officer.
   c) Becoming a detective.
   d) Becoming a doctor in medicine.

Chapter 2

3. After Evelyne Howard leaves Styles Court because of a discussion with Emily Inglethorpe she goes to live in Middlingham. What position does she find there?
   a) nanny
   b) housekeeper
   c) nurse
   d) pharmacist

4. What is the nickname of the colleague of Cynthia Murdoch?
   a) Nibs
   b) Buns
   c) Barny
   d) Snug

Chapter 3

5. Who wakes Mr. Hastings on the night of the murder?
   a) Nobody. He wakes up because of the noises.
   b) Lawrence Cavendish.
   c) John Cavendish.
   d) Dorcas.

Chapter 4

6. How does Alfred Inglethorpe explain that he did not sleep in his bed on the night of the murder?
   a) He didn't feel like it.
   b) He didn't want to wake his wife.
   c) Friends of his needed his help.
   d) He was drunk and slept with friends.

7. Why is the fact whether Emily Inglethorpe ate on the evening of her death of importance for the investigation?
   a) Her appetite says a lot about her emotional state.
   b) A big dinner could have slowed down the effect of the poison.
   c) A big dinner could have aggravated the effect of the poison.
   d) It is not important to the investigation.

8. Which of the following statements is not correct? When Poirot investigates the bedroom of Emily Inglethorp for the first time he..
   a) notices a coffee cup stamped to pieces.
   b) discovers a wet spot on the carpet.
   c) takes the documents out of the purple case.
   d) discovers the remains of a will in the ashes of the fireplace.

9. The sixth point of importance is…
   a) a piece of green fabric Poirot finds.
   b) That Dorcas overheard the argument between Mrs. and Mr. Inglethorpe
   c) the spare key to the purple box was missing
   d). An empty box of sleeping powders

10. What did Annie notice was strange about the cocoa?
    a) It was not heated yet.
    b) There was some kitchen salt on the tray.
    c) The tray was displaced by somebody else.
    d) The cocoa was brought up later than usual.

Chapter 5

11. How many coffee cups does Poirot count in the drawing room?
    a) 3
    b) 5
    c) 6
    d) 7

12. What is true about the last intact will of Mrs. Inglethorpe?
    a) She left her entire fortune to John Cavendish.
    b) She left her entire fortune to Alfred Inglethorp.
    c) She left her entire fortune to Evelyne Howard.
    d) She left her entire fortune to Lawrence Cavendish.

13. What does Lawrence think is the cause of dead of his stepmother?
    a) She was poisoned.
    b) Tetanus.
    c) Heart failure.
    d) Old age.

14. Who admits to not being a good friend to Mr. Hastings?
    a) John Cavendish.
    b) Hercule Poirot.
    c) Lawrence Cavendish.
    d) Mary Cavendish.

15. What element does Poirot thinks is implicating Alfred Inglethorpe in the murder?
    a) That he was not in the house at the moment of the murder.
    b) That the door of Mrs. Inglethorpes bedroom was closed from the inside so she must have opened the door for her husband.
    c) That everybody else suspects him.
    d) That Alfred Inglethorpe had an argument with Emily Inglethorpe on the day of her murder.

Chapter 6

16. Why is it impossible according to Dr. Wilkes that Emily Inglethorpe was poisoned by her medicine?
    a) Because it did not contain any poisonous substances.
    b) Because Emily Inglethorpe was too smart to take an overdose.
    c) Because the dose of strychnine in the medicine is too small to poison somebody.
    d) Because Emily Inglethorpe did not take the medicine that night.

17. How does Alfred Inglethorpe explain the last words of his wife?
    a) She was accusing him of her death.
    b) She was trying to tell him something.
    c) She wrongfully thought that Dr. Bauerstein was Mr. Inglethorpe.
    d) She was talking nonsense.

Chapter 7

18. What two things are of primary significance to Poirot about the day of the murder?
    a) The temperature and the green fabric he found.
    b) The temperature and the distinctive appearance of Alfred Inglethorpe.
    c) The temperature and the broken cup
    d) The broken cup and the distinctive appearance of Inglethorpe

19. Why is Poirot determined that Alfred Inglethorpe shall not be arrested for the murder of Emily Inglethorpe?
    a) Alfred will be found innocent.
    b) Alfred is innocent.
    c) There is not enough evidence to arrest Mr. Inglethorpe.
    d) The real murderer will escape.

Chapter 8

20. What is the reason that Alfred Inglethorp could not give a believable alibi for the time the strychnine was bought in the pharmacy?
    a) He was scared to admit he had been with Mrs. Raikes.
    b) He does not have an alibi.
    c) He was taking a walk and nobody could confirm this story.
    d) He wanted to be arrested.

21. Who inherits 'Styles Court'?
    a) Alfred Inglethorpe.
    b) John Cavendish.
    c) Lawrence Cavendish.
    d) Nobody.

22. What statement is true
    a) Annie shows Poirot where to find the dressing-up trunk
    b) Poirot finds the green dress he was looking for and a fake beard in the dressing-up trunk.
    c) Poirot finds the green dress he was looking for in the dressing-up trunk.
    d) Poirot finds a fake beard when looking for a green dress.

23. Which person does Evelyne Howard's intuition tells her committed the murder?
    a) Cynthia Murdoch
    b) Alfred Inglethorp

c) She does not say whom.
d) Mr. Hastings

Chapter 9

24. Where does Mr. Hastings hear Mary and John Cavendish arguing?

    a) The park
    b) Their bedroom
    c) In the village
    d) In the drawing room

25. When Cynthia Murdoch confides in Mr. Hastings, she tells him that certain people in the household hate her. Who is she talking about?

    a) John and Lawrence Cavendish
    b) Emily Howard and Mary Cavendish
    c) Lawrence Cavendish and Emily Howard
    d) Lawrence and Mary Cavendish

Chapter 10

26. The mysterious Dr. Bauerstein, although he does not turn out to be the murderer, nonetheless does turn out to be a criminal. What is his crime?
    a) Espionage
    b) Burglary
    c) Embezzlement
    d) Blackmail

27. Which of the following is true of Mary Cavendish?
    a) Her first husband died in prison.
    b) Her father died under mysterious circumstances.
    c) Her mother was Russian.
    d) Her father was shot for being a traitor.

28. The fourth letter that Mrs. Inglethorpe had sent on the day before the murder was addressed to
    a) Evelyne Howard.
    b) Alfred Inglethorp.
    c) A French music publisher.
    d) A nurse in Middlingham.

29. Whose fingerprints were found on the bottle of strychnine in the pharmacy?
    a) Lawrence Cavendish.
    b) Lawrence Cavendish and Cynthia Murdoch.
    c) Lawrence Cavendish and John Cavendish.
    d) Cynthia Murdoch.

Chapter 11

30. In the case against John Cavendish, what contention of the prosecution will Poirot be able to refute?
    a) That it was John who bought the strychnine in the pharmacy.
    b) That John would benefit from the death of Emily Inglethorpe.
    c) That John brought the coffee to Mrs. Inglethorpe's room.

d) That John was the one who burned the will.

31. In the court case against John Cavendish, who else is the prosecution trying to implicate in the murder.
    a) Mary
    b) Evelyne
    c) Lawrence
    d) Hastings

32. How does the defense explain the bottle of poison in John Cavendish's room?
    a) Somebody is trying to frame John Cavendish
    b) He used this to poison a stray dog.
    c) John hid this bottle for somebody else he is trying to protect.
    d) There was no bottle of poison in John's room.

33. Which of the following statements is false?
    a) John Cavendish claims that he does not remember the exact words that were used by his stepmother in their discussion.
    b) The prosecution believes that John bought the strychnine in the pharmacy.
    c) There is a consensus in the courtroom that the handwriting on the poison list in the pharmacy is Alfred Inglethorp's.
    d) Initially Lawrence denies that he touched the bottle of strychnine in the pharmacy.

Chapter 12

34. In addition to the strychnine, Emily Inglethorpe (along with Cynthia) was given a mild narcotic. Who drugged Emily Inglethorpe and Cynthia Murdoch?
    a) Dr. Bauerstein
    b) Mary Cavendish
    c) John Cavendish
    d) Evelyne Howard

35. Christie's knowledge of poisons would never be put to better use than in this novel. How was Emily Inglethorpe poisoned?
    a) Bromide was added to her medication.
    b) Strychnine was added to her coffee.
    c) Strychnine was added to the bromide powders.
    d) Strychnine was added to her medication.

Chapter 13

36. Which of the following is true of Evelyn Howard and Alfred Inglethorp's relationship?
    a) They hate each other intensely.
    b) They are trying to frame each other for murder.
    c) They suspect each other of committing the murder.
    d) They are cousins.

37. Whom does Evelyn Howard attempt to implicate in the murder?
    a) Mary Cavendish
    b) First Alfred Inglethorpe, then John Cavendish
    c) Alfred Inglethorpe
    d) John Cavendish

38. How does Poirot realize that there is a letter in the vase?
    a) There was one vase more on the mantle the first time he entered the room.

b) The vases were empty first.
c) He had to straighten the objects on the mantle twice.
d) The objects on the mantle were straight while they were crooked first.

39. Why did Poirot want John Cavendish to go on trial for the murder of Emily Inglethorpe?
a) He believed John was guilty.
b) John was interfering with the investigation.
c) He believed it would cause the real killer to confess.
d) He thought it would bring him and his wife closer together.